EX 7: IMAGE CAPTIONING USING CNN-RNN ARCHITECTURE

AIM

To generate meaningful captions for images using an encoder-decoder architecture that combines a pretrained Vision Transformer (ViT) for image feature extraction with GPT-2 for text generation.

PROCEDURE

1. **Import the required libraries, including PyTorch, Transformers, and PIL.**
2. **Load the pretrained ViT-GPT-2 image captioning model, image processor, and tokenizer.**
3. **Select the execution device (GPU or CPU) and set the model to evaluation mode.**
4. **Define a function to load an image and preprocess it into tensors.**
5. **Generate an image caption using beam search decoding.**
6. **Display the generated caption for the input image.**

CODE

```
import torch
from PIL import Image
from transformers import (
   VisionEncoderDecoderModel,
   ViTImageProcessor,
   AutoTokenizer
)

# Load the pretrained image captioning model
model_name = "nlpconnect/vit-gpt2-image-captioning"

model = VisionEncoderDecoderModel.from_pretrained(model_name)
processor = ViTImageProcessor.from_pretrained(model_name)
tokenizer = AutoTokenizer.from_pretrained(model_name)

# Select device
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model.to(device)
model.eval()
```

```
# Function to generate image captions
def generate_caption(image_path, max_length=30, num_beams=4):
    image = Image.open(image_path).convert("RGB")

    pixel_values = processor(
        images=image,
        return_tensors="pt"
    ).pixel_values.to(device)

    with torch.no_grad():
        output_ids = model.generate(
            pixel_values,
            max_length=max_length,
            num_beams=num_beams,
            early_stopping=True
        )

    caption = tokenizer.decode(
        output_ids[0],
        skip_special_tokens=True
    )

    return caption


# Load an image and generate its caption
if __name__ == "__main__":
    img_path = "download.jpeg"  # Replace with your image filename

    caption = generate_caption(img_path)
    print("Generated Caption:", caption)
```

OUTPUT

Input Image



Caption: a black and white dog sitting in a grassy field



RESULT

Thus, image captioning was performed successfully using a pretrained Vision Transformer (ViT) encoder and GPT-2 decoder. Beam search was used to generate a meaningful textual description of the input image.